

THERMACART · REQUIREMENTS

ThermaCart requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Status from TCT-CAL-001 v0.5: R9 met by design with the frame lip; R14 and R16 figures updated; no status category changed
0.6	2026-10-02	Amish Chadha	R9 status note: the low lip decided on 2026-10-02 (TCT-DDR-003 A1 (b)) is still to be designed
0.5	2026-10-01	Amish Chadha	Status from TCT-CAL-001 v0.3 for the constructable design (TCT-DDR-003); R16 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from TCT-CAL-001; R12 oven temperature corrected to the RT 70 HC limit; R16 budget scope per TCT-DDR-001 D6
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

ThermaCart requirements

These are the requirements for the TC-L cartridge and its adapter frame. The targets were proposed at TRL 2, and the design choices behind them were decided by Amish on 2026-09-25 (TCT-DDR-001 and TCT-DDR-002). Under TCT-DDR-002 the C25 band of R1 is widened to 22 to 27 °C (O9), R5 is relaxed to 38 % for the stock-tube build with 50 % kept as the target for a later extrusion (O4), and R14 carries a labelled handling rule for H70 cartridges taken from an oven (O7). R10 keeps its 8 h target (O5). The status column comes from TCT-CAL-001 v0.5, which uses the decided dark finish (O8) and the constructable design of TCT-DDR-003. One requirement is **not met** on paper (R10), one is **over its value-engineering target** (R16; `budget_usd` is a hypothetical control target, not a limit), one is **at risk** (R12), three can only be settled by tests (R6, R7 and R17) and twelve are met. No requirement text changed in v0.5 or v0.7.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (TCT-CAL-001)
R1	Offer a small set of standard temperature grades	At least three: C5 (melt 4 to 6 °C, cold chain), C25 (22 to 27 °C, comfort; widened from 24 to 27 °C by TCT-DDR-002, 09), H70 (68 to 72 °C, hot holding)	Material datasheets; later DSC test of each fill batch	Met: RT 5 HC 5 to 6 °C, RT 25 HC 22 to 26 °C, RT 70 HC 69 to 71 °C
R2	Common envelope that fits kitchen and cooler formats	Fits a GN 1/3 slot: at most 325 x 176 mm footprint and 65 mm high, handle included	Model and drawing	Met: 323.4 x 152.4 x 63.5 mm with the bail stowed (TCT-DWG-001)
R3	Enough stored energy per cartridge	60 Wh (216 kJ) or more of usable storage within ± 3 K of the melt point	Calculation per grade	Met: C5 65.4 Wh, C25 60.5 Wh (thin margin), H70 72.9 Wh
R4	Carry with one hand	Filled mass 3.0 kg or less	Mass estimate, then weighing	Met, thin margin: 2.90 to 2.98 kg
R5	Efficient use of mass	PCM at least 38 % of filled mass for the stock-tube build; 50 % remains the target for a later custom extrusion (relaxed by TCT-DDR-002, 04)	Mass estimate	Met, thin margin: 38.1 to 39.6 %
R6	Sealed for life	No leak through 1,000 thermal cycles and a 1 m drop onto concrete in any state	Seal design review at TRL 3; cycle and drop tests belong to TRL 4	Not verifiable at TRL 3; an end-on drop of a molten cartridge could see a surge of up to 5 bar
R7	Long life	1,000 cycles or more with less than 10 % loss of latent capacity	Literature and supplier data at TRL 3	Not verifiable at TRL 3; plausible for paraffins
R8	No pressure hazard	Internal gauge pressure 0.5 bar or less at the highest charge temperature; ullage of at least 10 % of the inner volume at that temperature	Calculation	Met: 0.00 bar at the fill temperature, +0.29 bar at 20 K above the limit; wall stress 56 MPa under the cold vacuum
R9	Wrong grade cannot be fitted	Each grade has its own nose key; a frame for one grade rejects the others	Design review of key geometry	Met by design: every key pairing checked, and the frame's low lip (TCT-DEC-001, item 1) holds a wrong grade out, resting on the lip 4 mm high and 8 mm short of seated; checked in the model with the bail stowed and raised (TCT-CAL-001, G2 and G3)
R10	Cold-chain hold time	Two C5 cartridges keep a 25 L cooler (heat leak about 0.5 W/K, estimate) at 2 to 8 °C for 8 h or more at 32 °C ambient	Heat balance calculation	Not met: 2.4 h with the decided dark finish (1.9 h bare); the stored energy would last 9.7 h, but air-side heat transfer limits the hold. Produce laid

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (TCT-CAL-001)
				directly on the cartridges (TCT-DDR-002, O5) is not credited and waits for a TRL 4 measurement, which is on hold
R11	Hot-holding time	One H70 cartridge keeps food in an insulated GN carrier (about 0.25 W/K, estimate) at 63 °C or more for 4 h or more at 25 °C ambient	Heat balance calculation	Met: 13.6 h with the dark finish (7.2 h bare)
R12	Recharge with common equipment	Full recharge in 8 h or less: C5 and C25 in a domestic freezer or refrigerator, H70 in an oven set to 85 °C (never above 90 °C) or on a 60 W heating pad	Calculation (Stefan problem plus fin convection)	At risk: with the dark finish, C5 freezer 4.2 h, C25 refrigerator 4.6 h, H70 oven 5.3 h; H70 pad 2.7 to 10.3 h
R13	State of charge visible	Charged or melted state readable at a glance without tools	Design review	Met by design (clear melt indicator tube holding the same PCM, TCT-DDR-003)
R14	Safe to handle when charged	Handle touch temperature below the ISO 13732-1 burn threshold for 10 s contact on every grade; for an H70 cartridge taken from an oven, the label states a handling rule (oven gloves, or wait 12 min) and pad charging is the preferred method (TCT-DDR-002, O7)	Calculation of handle temperature; label review	Met with the handling rule: bail on its thermal break (phenolic washers under the lug angles and the bolt heads) settles at 39.2 °C on pad charging; after an oven it is at 85 °C and falls below 55 °C in about 10 min, which the 12 min label rule covers
R15	Materials compatible with each fill	No corrosion, swelling or permeation: aluminium shell with FKM or NBR seals for paraffins; coated shell or polymer liner for salt hydrates	Compatibility review against supplier data	Met by design for paraffin grades (epoxy rated 120 °C or more); open for salt hydrates
R16	Low cost and garage-buildable	\$50 or less in parts per cartridge and \$20 or less per frame; stock sections, drill press and taps only. The \$80 value-engineering target (budget_usd , a hypothetical control target) covers one C5 cartridge and one frame (TCT-DDR-001 D6)	Priced BOM	Over the value-engineering target by \$32.50 per cartridge: \$82.50 with the dark finish, the parts added for construction and the decided grade band and side marking; frame \$17.50, \$2.50 under its target; one cartridge and one frame \$100, \$20 over the \$80 target (indicative prices; quotes wait with TRL 4, TCT-DDR-002, O6)
R17	Open and documented	Envelope, key geometry, grade table and fill recipe published under CERN-OHL-S-2.0	Repository review	Not verifiable at TRL 3; envelope, key and grade tables are published, the fill recipe waits for TRL 4
R18	Recoverable at end of life	Fill can be drained or melted out through the fill port and the shell reused; fill named on the label	Design review	Met by design

■ Assumptions

- PCM data from the Rubitherm datasheets for RT 5 HC, RT 25 HC and RT 70 HC, taken at the low side of their ± 7.5 % tolerance; usable storage is the latent heat plus sensible heat within ± 3 K of the melt point (TCT-CAL-001, Table 1).
- Fill of 1.10 to 1.18 kg: 90 % of a 1.655 L inner volume, filled as liquid at each grade's maximum operating temperature (45, 65 and 90 °C).
- Hot-holding threshold of 63 °C follows common food-safety practice (the US FDA Food Code uses 57 °C, 135 °F); the lower value would lengthen the hold time.
- Cooler and carrier heat leaks are estimates for typical polyurethane and EPP boxes and must be replaced with measured values from a partner's real boxes.
- The burn thresholds used for R14 (about 55 °C for bare metal and 70 °C for plastics, 10 s contact) are commonly quoted from ISO 13732-1 and have not been checked against the standard.